

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

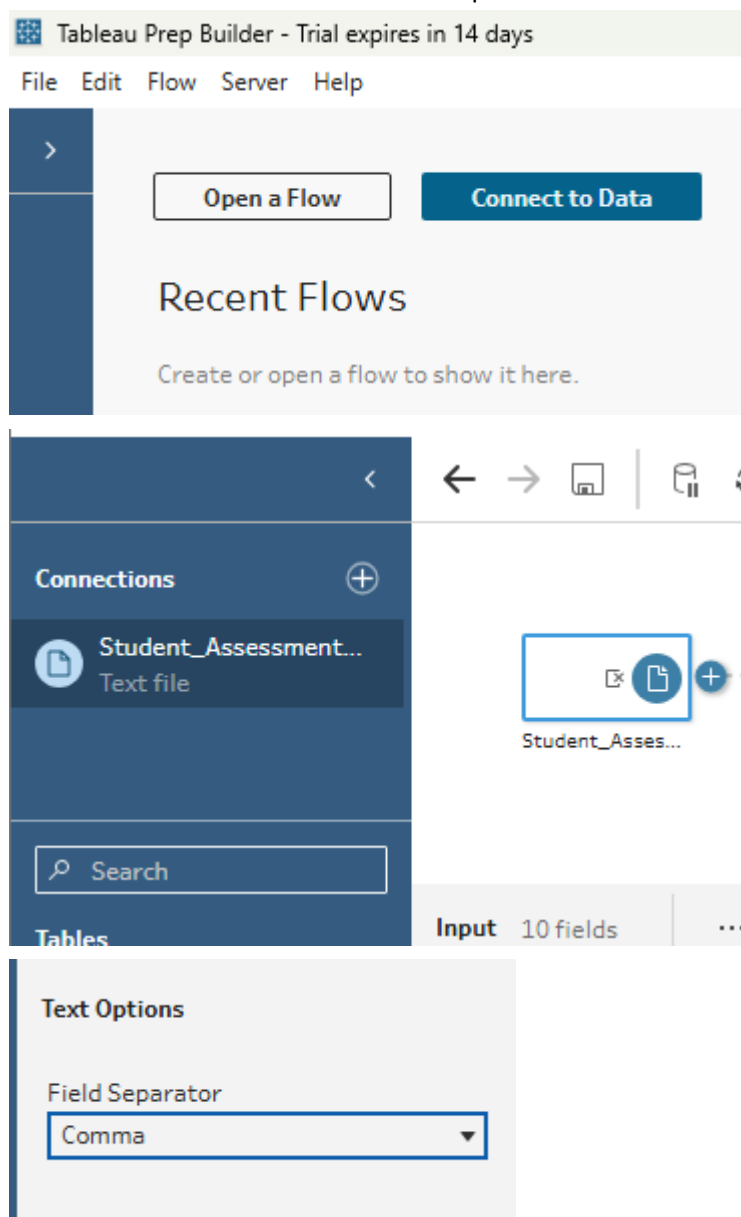
Practical – X

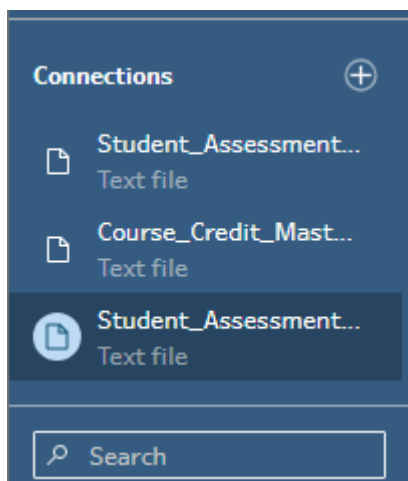
Roll No.: T038	Name: Akansha Pritam Ranade
Class: TYIT	Batch: 1
Date of Assignment: 03-09-2026	Date/Time of Submission: 04-09-2026

Part B – Program

Data Preparation and Transformation using Tableau Prep

- a) Connect and load data into Tableau Prep





Step 4: Add an Initial Clean Step to Term 1

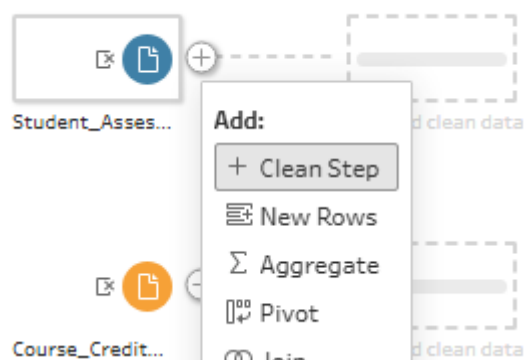


Diagram illustrating data flow:

- Student_Asses...** (Data Source) → **Clean 1** (Data Operation)
- Course Credit** (Data Source) → **View and clean data** (Data Operation)

The **Clean 1** operation is shown in a table view with 9 fields and 16 rows. The table displays data for **Student_ID**, **Student_Name**, and **Department**.

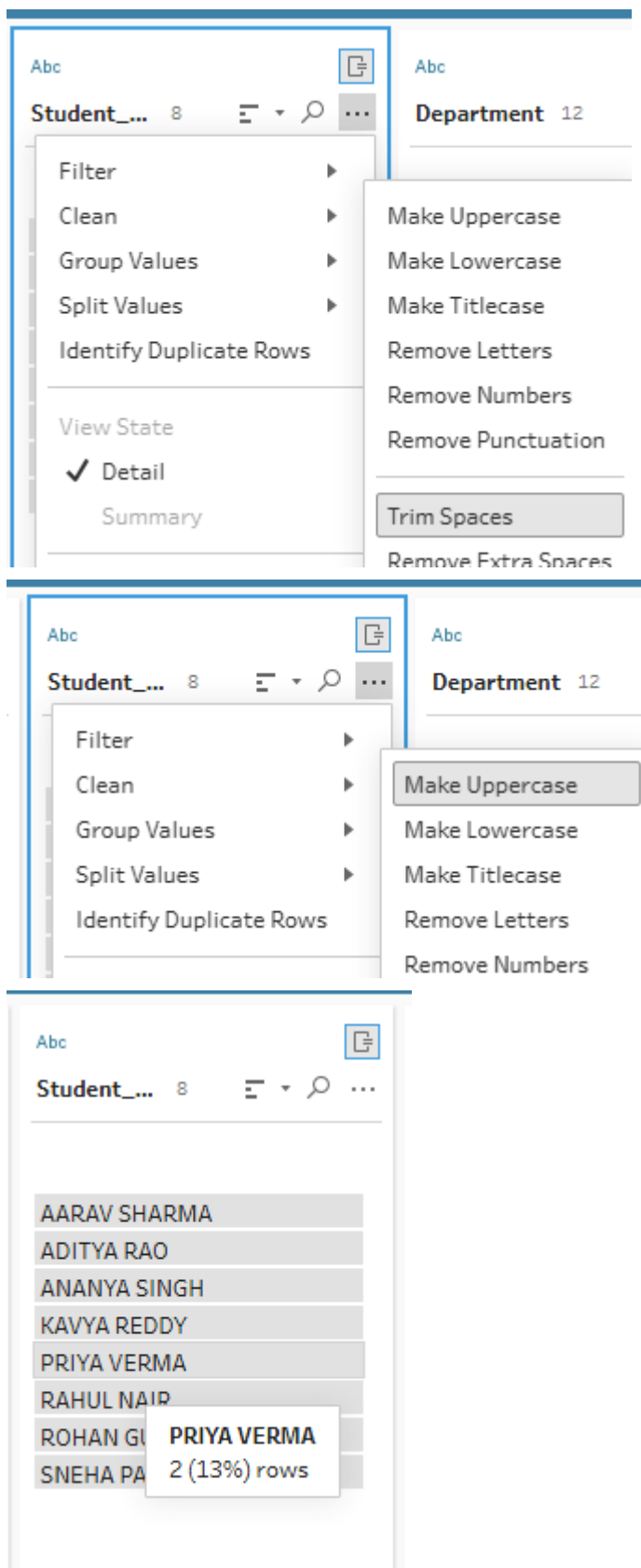
Student_ID	Student_Name	Department
S101	Aarav Sharma	Comp Sci
S102	aditya Rao	Computer Science
S103	Ananya Singh	CS
S104	Kavya Reddy	CSE
S105	priya VERMA	ECE
S106	Rahul Nair	Electronics

b) Clean and profile data

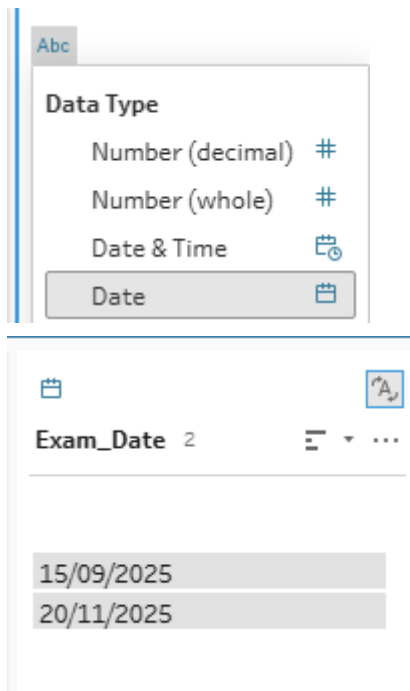
The **Clean 1** operation is shown in a table view with 9 fields and 16 rows. The table displays data for **Assessm...**, **Exam_Date**, and **Subject_Code**.

Assessm...	Exam_Date	Subject_Code
Final	15-09-2025	CS101
Midterm	20-11-2025	EC101
	2025.09.15	IT101
	2025.11.20	ME101
	2025/09/15	
	2025/11/20	

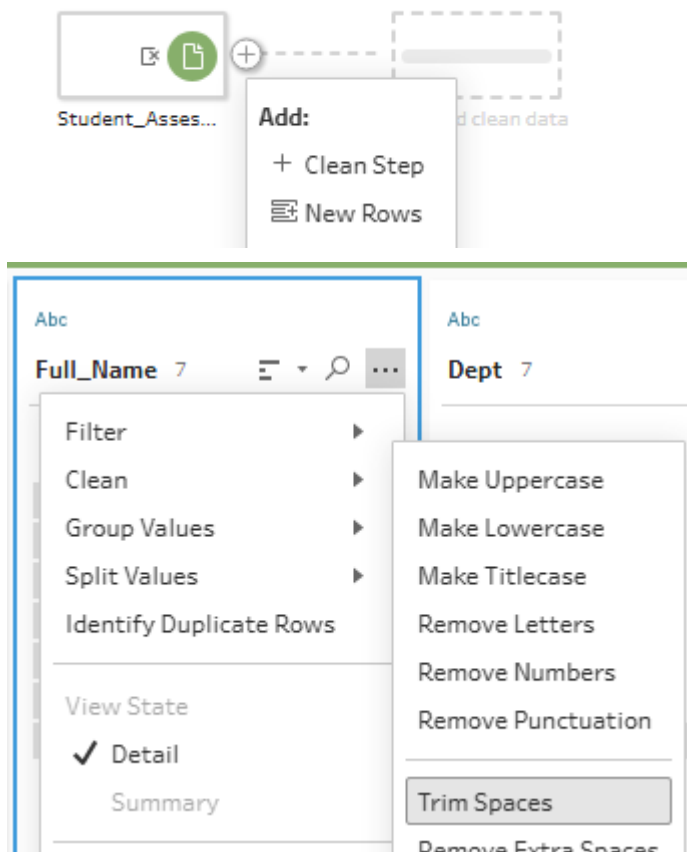
Step 2: Clean Whitespace and Standardize Letter Casing



Step 3: Convert Exam_Date to Date Data Type



Step 4: Clean the Second Dataset (Student_Assessments_Term2.csv)



Abc

Full_Name 7

Aarav Sharma
Ananya Singh
Kavya Reddy
Priya Verma
Rahul Nair
Rohan Gupta
Sneha Patel

Assessment_... 2

10/02/2026
25/04/2026

Step 5: Review the Changes Pane

Clean 2 9 fields 14 rows

Changes (1)

Trim Spaces

Full_Name

Spaces trimmed from all values

c) Group and standardize values

Step 1: Inspect the Department Field in Term 1 (Clean 1)

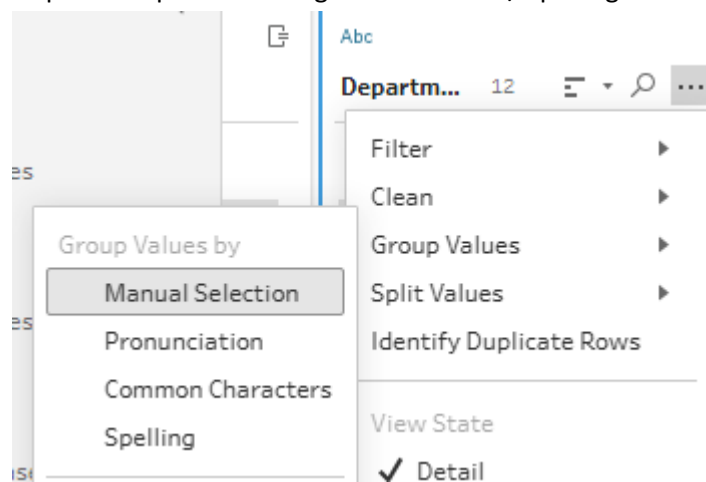


Abc

Department 12

Comp Sci
Computer Science
CS
CSE
ECE
Electronics
Electronics Engg
Info Tech
Information Tech
IT
Mech Engg
Mechanical

Step 2: Group Values Using Pronunciation / Spelling or Manual Selection



Group Values by Manual Selection

Department 12

Comp Sci

Computer Science

CS

CSE

ECE

Electronics

Electronics Engg

Info Tech

Information Tech

IT

Mech Engg

Mechanical

Group Values by Manual Selection

Department 9

Comp Sci

ECE

Electronics

Electronics Engg

Info Tech

Information Tech

IT

Mech Engg

Mechanical

Comp Sci 4 members

☒ Comp Sci
 ☒ Computer Science
 ☒ CS
 ☒ CSE

☐ ECE
 ☐ Electronics
 ☐ Electronics Engg
 ☐ Info Tech
 ☐ Information Tech
 ☐ IT
 ☐ Mech Engg
 ☐ Mechanical

Group Values by Manual Selection

Depa... 7

Comp Sci

Electronics

Info Tech

Information Tech

IT

Mech Engg

Electronics 3 members

☒ ECE
 ☒ Electronics
 ☒ Electronics Engg

☐ Comp Sci
 ☐ Computer Science
 ☐ CS
 ☐ CSE

Group Values by Manual Selection

Department 5	Information Tech 3 members
☐ Comp Sci	☒ Info Tech
☐ Electronics	☒ Information Tech
☒ Information Tech	☒ IT
☐ Mech Engg	☐ Comp Sci
☐ Mechanical	☐ Computer Science
	☐ CS
	☐ ---

Group Values by Manual Selection

Department 4	Mechanical 2 members
☐ Comp Sci	☒ Mech Engg
☐ Electronics	☒ Mechanical
☐ Information Tech	☐ Comp Sci
☐ Mechanical	☐ Computer Science
	☐ CS
	☐ CSE
	☐ ECE

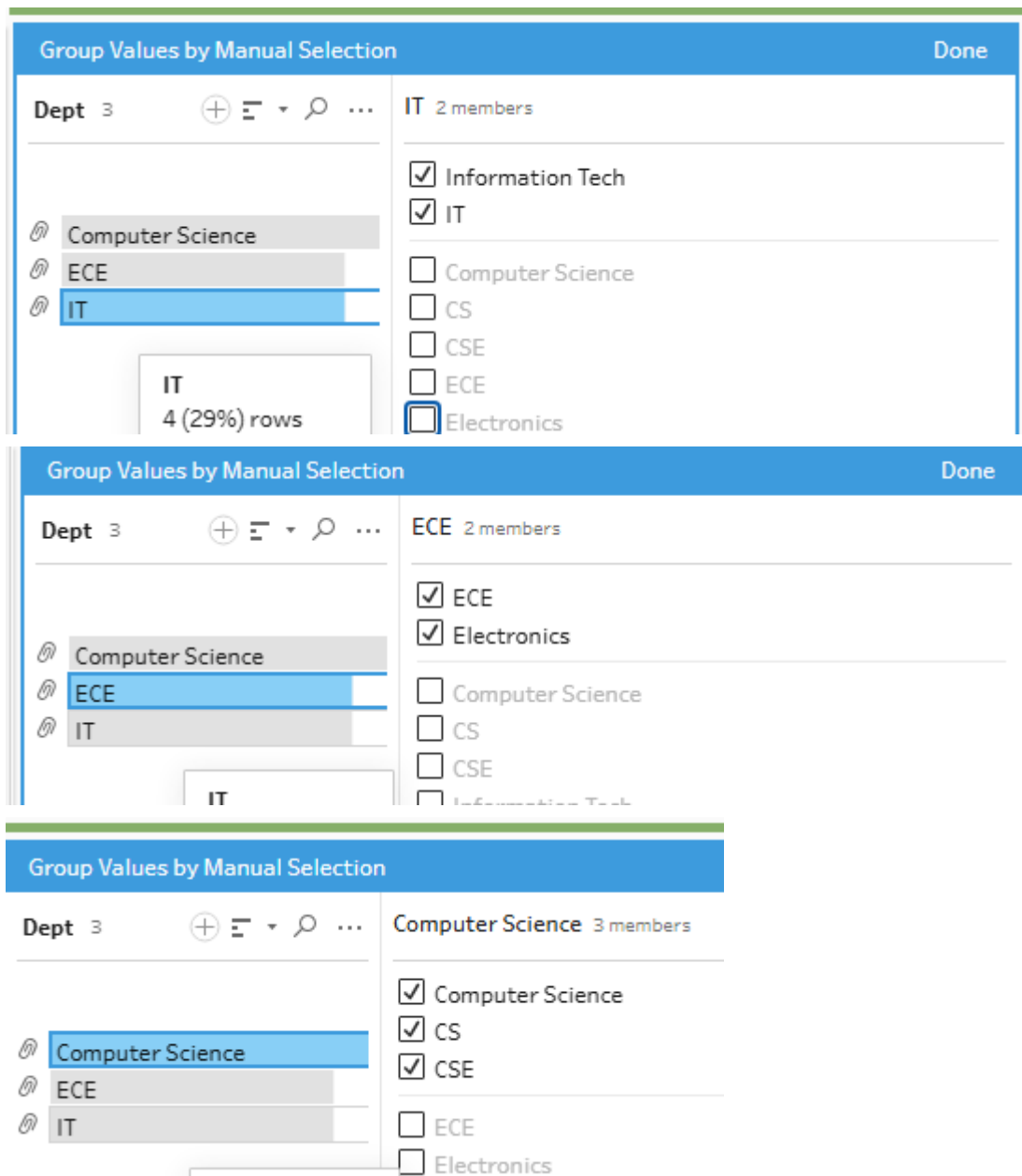
Abc 

Departm... 4    

☐ Comp Sci
☐ Electronics
☐ Information Tech
☒ Mechanical

Step 3: Repeat Grouping on Term 2 (Clean 2)

Dept 7	Exam_Type 2
Abc      Group Values by Manual Selection Pronunciation Common Characters	Abc Filter  Clean  Group Values  Split Values  Identify Duplicate Rows View State



d) Merge mismatched fields

Step 1: Create the Union Step on the Canvas



Step 2: Inspect Mismatched Fields in the Union Pane

Union 1 18 fields 30 rows

Select All Filter Values... Identify Duplicate Rows Create Calculated Field...

Settings Changes (0)

Inputs

- Clean 1
- Clean 2

Resulting Fields

16 Mismatching fields from 18 resulting fields.

Mismatched Fields

- Student_ID ☒
- Full_Name ☐
- Student_Name ☒
- Std_ID ☐
- Dept ☐
- Exam_Type ☐

Union Results ☐ Show only mismatched fields

Table Na...	Student_ID	Student_Name	Department
Student_Assessments_Term1	null	null	null
Student_Assessments_Term2	S101	AARAV SHARMA	Comp Sci
	S102	ADITYA RAO	Electronics
	S103	ANANYA SINGH	Information
	S104	KAVYA REDDY	Mechanical
	S105	PRIYA VERMA	
	S106	RAHUL NAIR	
	S107	ROHAN GUPTA	
	S108	SNEHA PATEL	

Step 3: Merge the Mismatched Fields

Abc

Student_ID 8

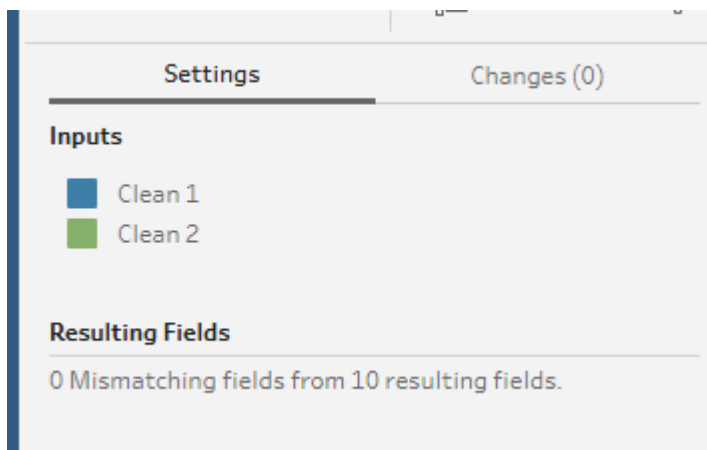
S101
S102
S103
S104
S105
S106
S107
S108

Abc

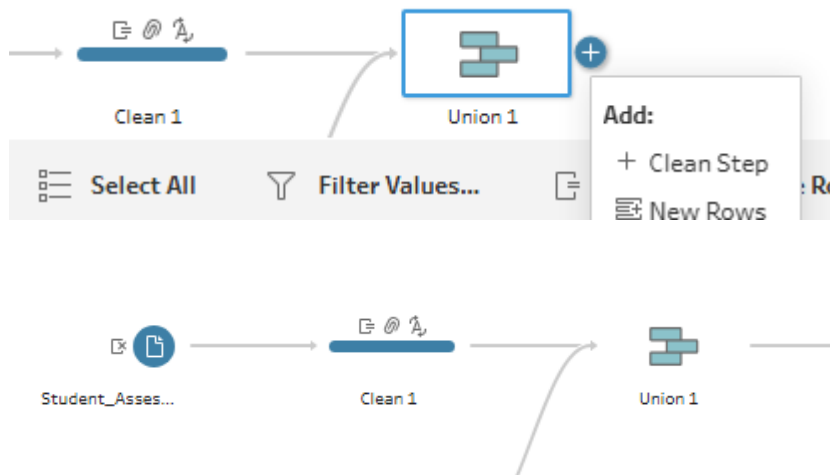
Student_... 15

Student_Name
15 unique values

Aarav Sharma
AARAV SHARMA
ADITYA RAO
Ananya Singh
ANANYA SINGH
Kavya Reddy
KAVYA REDDY
Priya Verma
PRIYA VERMA
Rahul Nair
RAHUL NAIR
Rohan Gupta



Step 4: Add a Clean Step Post-Union



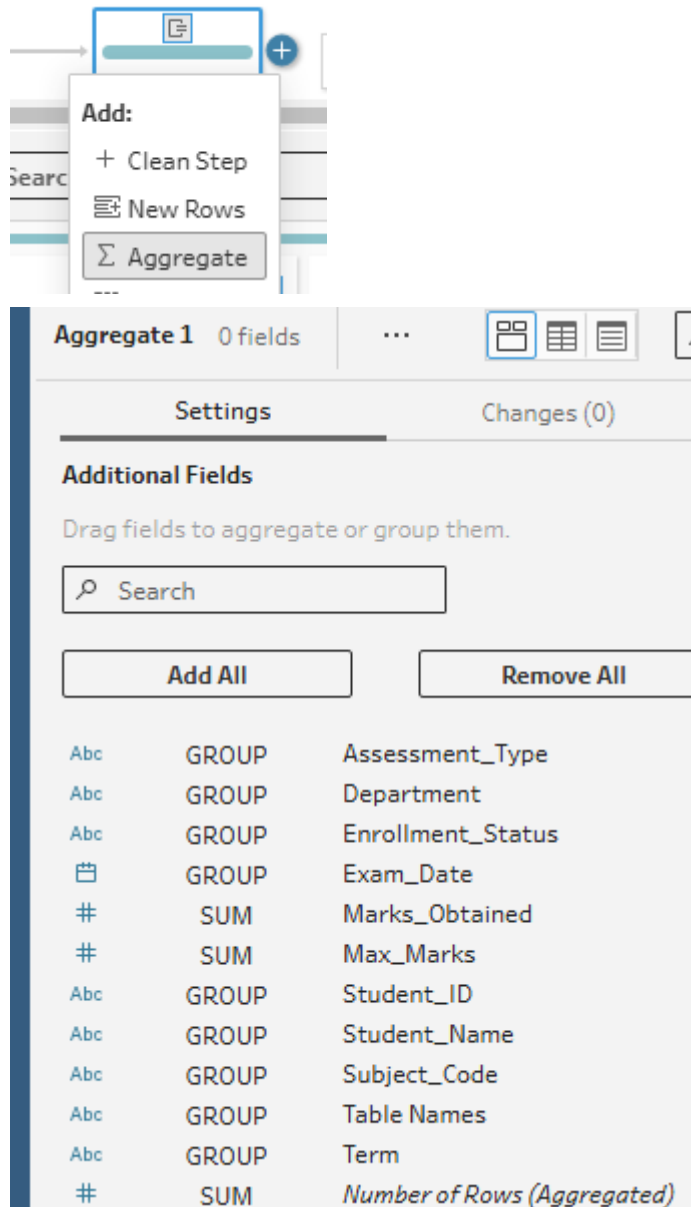
- e) Perform aggregations in Prep
Step 1: Add a Term Identifier Field in Clean 3

Field Name

Term

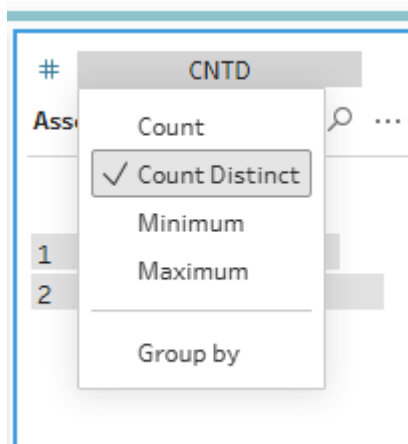
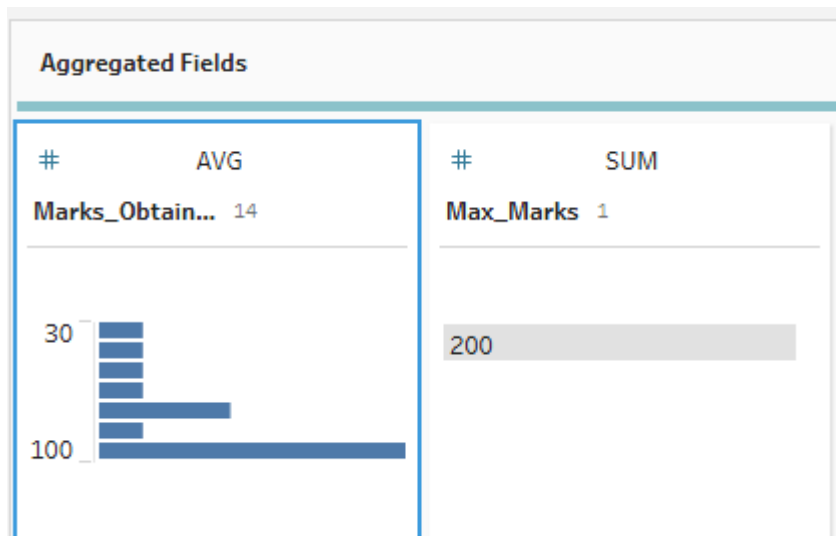
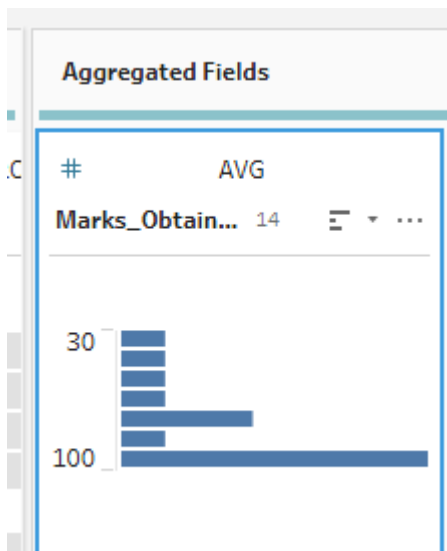
```
IF CONTAINS([Table Names], "Term1") THEN "Term 1"
ELSEIF CONTAINS([Table Names], "Term2") THEN "Term 2"
ELSE "Other"
END
```

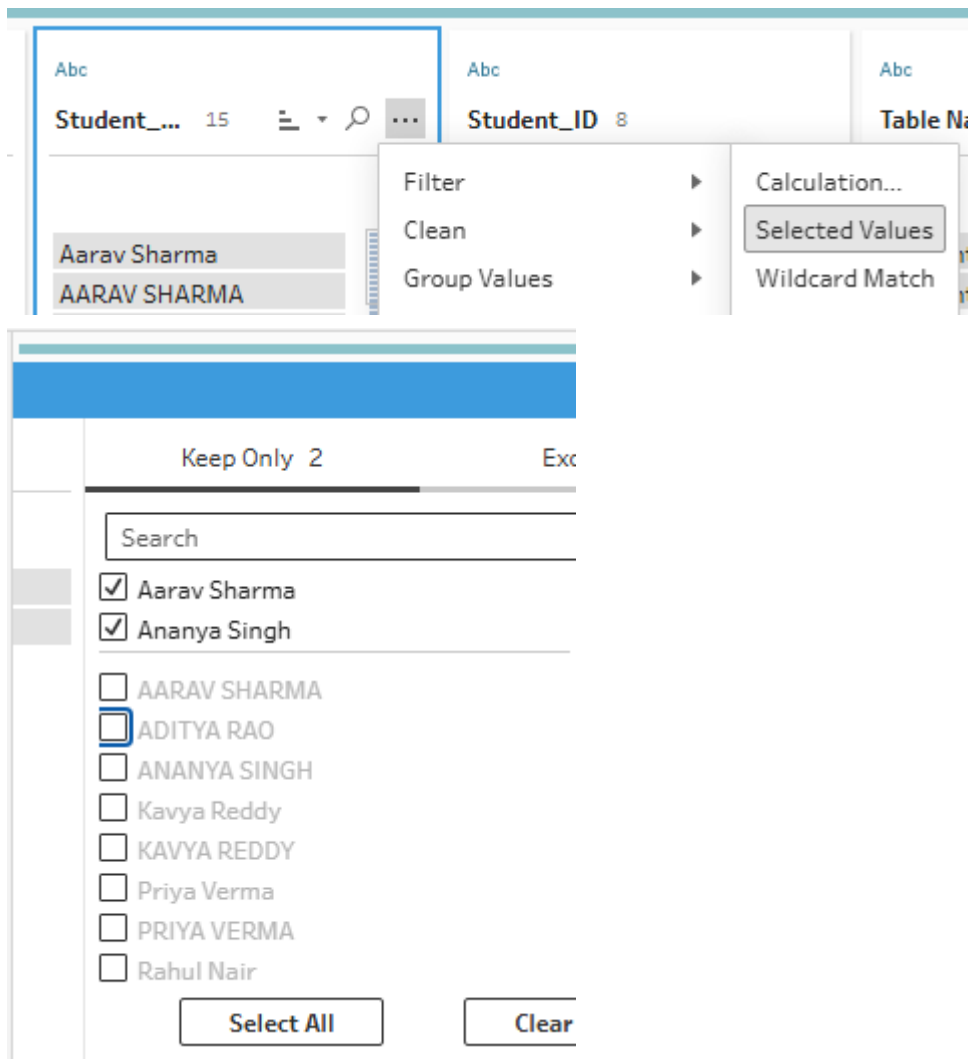
Step 2: Add an Aggregate Step to the Flow Canvas



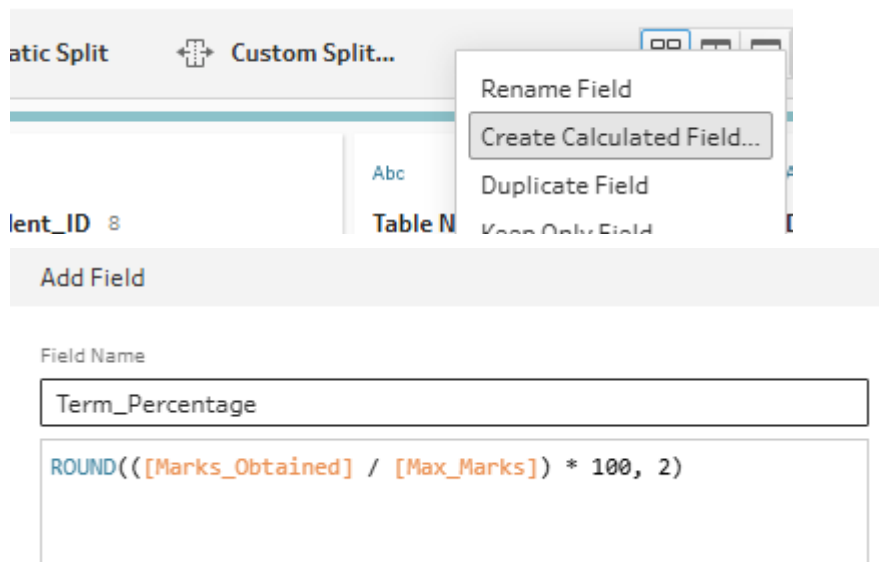
Step 3: Configure Grouped Fields (Dimensions)

1. Drag Student_ID into the Grouped Fields box.
2. Drag Student_Name into the Grouped Fields box.
3. Drag Department into the Grouped Fields box.
4. Drag Term into the Grouped Fields box.





Step 3: Create the Normalized Term Percentage Metric



Step 4: Create the Categorical Letter Grade Transformation

Field Name

Letter_Grade

```
IF [Term_Percentage] >= 85 THEN "A  
(Distinction)"  
ELSEIF [Term_Percentage] >= 75 THEN "B (First  
Class)"  
ELSEIF [Term_Percentage] >= 60 THEN "C (Second  
Class)"  
ELSEIF [Term_Percentage] >= 40 THEN "D (Pass  
Class)"  
ELSE "F (Fail / Re-appear)"  
END
```

Step 5: Create Academic Standing Compliance Flag

Field Name

Academic_Standing

```
IF [Term_Percentage] >= 60 THEN "Satisfactory"  
ELSE "Academic Probation (<60%)"  
END
```

Step 6: Verify Transformed Profile Distributions

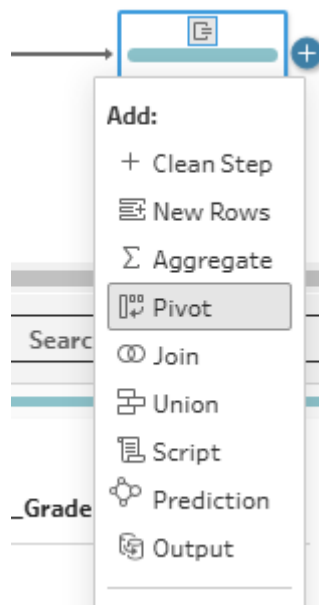
Clean 6 11 fields 15 rows ...

Changes (3)

- Calculated Field
Term_Percentage
ROUND(((Marks_Obtained]/
[Max_Marks]) * 100, 2)
- Calculated Field
Letter_Grade
IF [Term_Percentage] >= 85 THEN "A
(Distinction)" ELSEIF
- Calculated Field
Academic_Standing
IF [Term_Percentage] >= 60 THEN
"Satisfactory" ELSE "Academic

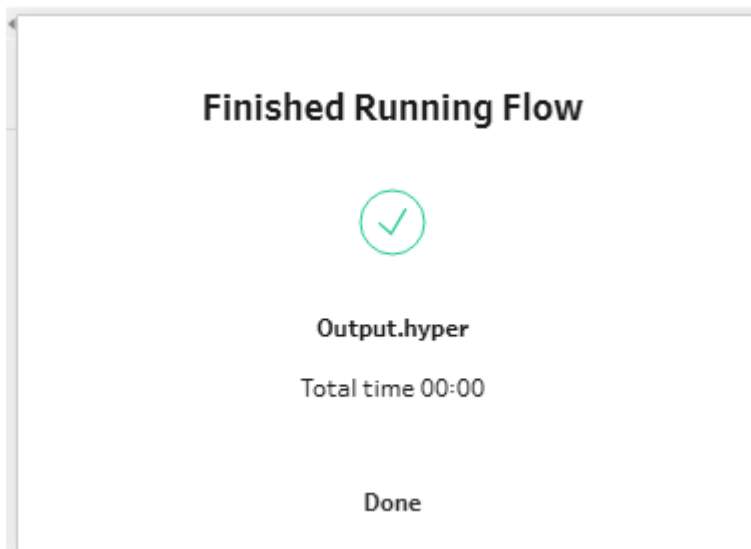
g) Create and automate data preparation flows

Step 1: Add an Output Step to the End of the Pipeline

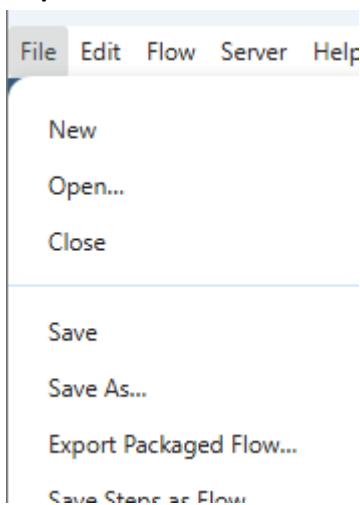


Step 2: Configure Output Settings in the Bottom Pane

A screenshot of the 'Output' settings pane in Tableau Prep Builder. The pane is titled 'Output 11 fields'. It contains several configuration options: 'Save output to' is a dropdown menu set to 'File'; there is a 'Browse' button below it; 'Name' is a text field set to 'Output'; 'Location' is a text field set to 'C:\Users\itlab\OneDrive\Documents\My Tableau Prep Repository\Datasources'; 'Output type' is a dropdown menu set to 'Tableau Data Extract (.hyper)'; 'Write Options' is a section with a description 'Select an option to create or update your output table.' and a dropdown menu set to 'Create table'. At the bottom right of the pane is a blue 'Run Flow' button.



Step 4: Save and Automate the Flow File



Part C – Curiosity Question

1. What would happen if the same value is written in different formats, such as "Mumbai" and "mumbai"?
2. Why is it important to identify missing or incorrect data before analysis?
3. How can grouping values help when similar categories have different names?
4. What happens when two datasets have fields with different names but contain the same type of information?
5. Why would you create a Tableau Prep flow instead of cleaning the data manually every time?